RONAK NARKHEDE

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EDUCATION

University of Minnesota | Master of Science in Robotics | *Dr. Timothy Kowalewski* St Paul, MN
GPA: 3.38/4.0 Sept 2024 – May 2026
SRM University | Bachelor of Technology in Mechatronics Chennai, India
GPA: 8.32/10.0 Sept 2020 – May 2024

TECHNICAL SKILLS

Programming Languages: C++ (14/17), Python, MATLAB
Software & Libraries: ROS 1; MoveIt; OMPL; OpenCV; MuJoCo; Gazebo; Eigen; Docker; Git
Featured Knowledge: Model-Free Adaptive Control; Jacobian Estimation; Kalman Filter; Teleoperation; Kinematics
Tools & Platforms: Docker, Linux, SolidWorks, Git, MuJoCo, Gazebo, Robot Programming (RoboDK, ABB RAPID, URScript)

PROFESSIONAL EXPERIENCE

Medical Robotics and Devices Lab | Graduate Research Assistant Minneapolis, MN | Sept 2024 – Present

- Built a closed-loop autonomous catheter navigation system using Electromagnetic tip tracking, applying online Jacobian estimation to solve inverse kinematics under kinematic uncertainty.
- Implemented a physics based guidewire digital twin in Nvidia Isaac Sim using a 135 DOF mass soring damper chain with pre-curved tip, torsion joints, and force-coupled CV tracking for phantom navigation.
- Developed a multi threaded, teleoperation platform for remote surgical robot control using FastAPI Webserver, WebSockets, RESTful APIs, and WebRTC streaming.
- Deployed the teleoperation platform for live demos and federal grant evaluations, maintaining 24/7 uptime. Handled downtime incidents, troubleshooting hardware software integration issues to get the system back up quickly.
- Built a safety stop feature using OpenCV and ArUco markers to track the guidewire and preventit from fully retracting out of the vascular phantom.
- Worked with neurosurgeon Dr. Andrew Grande, professor Dr. Karthik Desingh and Medtronic engineers throughout development, designing the control interface and test scenarios around their clinical feedback.

Choice Robotics Lab | Research Volunteer Minneapolis, MN | Jan 2025 – Present

- Developed robot learning algorithms for contact-rich manipulation on the ALOHA bimanual platform. Used both Behavioral Cloning, and Soft Actor-Critic RL to achieve 78% task success on Peg-in-Hole and Cube Transfer benchmarks.
- Designed simulation environments in MuJoCo and Gymnasium to rapidly prototype and evaluate control algorithms before deployment on physical hardware.

PROJECTS

UR5 Manipulation & Force Control | *UR5, URScript, Python, RoboDK* Fall 2025

- Programmed a UR5 robot for autonomous flashlight assembly using Python (RoboDK) and URScript, with pneumatic grippers via digital I/O and force controlled threading.
- Implemented Cartesian Control using raw force sensor feedback for compliant peg-in-hole insertion without any vision feedback.
- Calibrated system parameters like TCP offsets and Center of Gravity to resolve inverse kinematics singularities.

Franka 3 Integration | *Docker, C++, Real-time Control, Point Clouds* Fall 2025

- Contributed to lab bring up of Franka Research 3 arms by containerizing multiple real-time control methods in Docker and co-developing a C++ framework for unified access to arm state and depth camera data.

Vision Based Robot Control | *Stereo Camera, ArUco, ROS Noetic, Python* Fall 2024

- Designed a 7 node ROS Noetic pub/sub architecture for a wireless robot teach pendant, with nodes for stereo camera, ArUco detection, coordinate transformation, Levenberg-Marquardt IK solving, RViz simulation, and robot execution, communicating over custom topics using geometry_msgs and rosserial for ESP32 WLAN integration.

PUBLICATIONS

Towards Remote Thrombectomy with Telerobotically-Driven Guidewires *Accepted*
(Lead co-author) | DOI: DMD2026-1079 | *2026 Design of Medical Devices Conference* April 2026

Estimating the Non-parametric Jacobian of a Tendon-driven Soft Robot *Accepted*
DOI: DMD2026-1076 | *2026 Design of Medical Devices Conference* April 2026